Formal Logic 1

Christopher R. Walker

September 29, 2026

1. **What Is Logic?**

   Required Reading: Magnus et al., 2025, Chapters 1-5 (pp. 2-48)
   Digging Deeper: Smith, 2020, Chapters 1-4, 6–7, 11 (pp. 1-39, 44–58, 94-104)

   Focus: informal presentation of *validity*, *soundness*, *entailment*, *satisfiability*; distinguishing *deductive* from *inductive* arguments; translating ordinary English into truth-functional logic (TFL); understanding the meaning of TFL connectives;

2. **Truth and Grammar in TFL**

   Required Reading: Magnus et al., 2025, Chapters 6-11 (pp. 49-84)
   Digging Deeper: Smith, 2020, Chapters 8-13, 18-19 (pp. 59-120, 148-172)

   Focus: *sentences* in TFL, *parse trees*, connective *scope*, types of *ambiguity*; understanding *truth-functional* connectives; building *truth tables* for complex sentences;

3. **Semantic Strategies in TFL**

   Required Reading: Magnus et al., 2025, Chapters 12-15 (pp. 85-108)
   Digging Deeper: Smith, 2020, Chapters 14-16 (pp. 120-137)

   Focus: testing logical properties with truth tables; efficiency strategies when using truth tables; using partial tables for validation;

4. **Natural Deduction in TFL**

   Required Reading: Magnus et al., 2025, Chapter 16-17 (pp. 110-141)
   Digging Deeper: Smith, 2020, Chapters 20-22 (pp. 172-211)

   Focus: presentation of the Fitch-style *proof* system; *introduction* and *elimination* rules for TFL connectives

5. **Proofs in TFL**

   Required Reading: Magnus et al., 2025, Chapter 18 (pp. 142-161)
   Digging Deeper: Smith, 2020, Chapter 17 (pp. 145-147)

   Focus: proof strategies; working backward from the conclusion; working forward from the premises; building and exploiting *subproofs*

6. **Additional Rules**

   Required Reading: Magnus et al., 2025, Chapters 19-21 (pp. 162-181)
   Digging Deeper: Smith, 2020, Chapter 23 (pp. 211-215)

   Focus: extending the basic system with additional rules;

7. **Basic TFL Metalogic**

   Required Reading: Magnus et al., 2025, Chapter 22 (pp. 182-188)
   Digging Deeper: Smith, 2020, Chapter 24 (pp. 216-228)

   Focus: connecting *proof theory* to the semantics of *truth-functional* logic;

8. **An Informal Survery of First-Order Logic**

   Required Reading: Magnus et al., 2025, Chapter 23-27 (pp. 190-242)
   Digging Deeper: Smith, 2020, Chapters 25-27, 31 (pp. 228-258, 293-298)

   Focus: translating simple *universal* and *existential* sentences; *predicate* notation; nested *quantifiers*; the *identity* predicate; *definite descriptions*;

9. **Grammar of FOL**

   Required Reading: Magnus et al., 2025, Chapter 28-29 (pp. 243-256)
   Digging Deeper: Smith, 2020, Chapters 28-30 (pp. 258-289)

   Focus: *sentences* and *scope ambiguities* in first-order logic;

10. **FOL Semantics**

    Required Reading: Magnus et al., 2025, Chapter 30-35 (pp. 258-295)

    Focus: assessing the truth of FOL sentences relative to an *interpretation*; testing validity via *counter-models*

11. **FOL Proofs**

    Required Reading: Magnus et al., 2025, Chapters 36-41 (pp. 297-335)
    Digging Deeper: Smith, 2020, Chapters 32-42, Appendix (pp. 299-412)

    Focus: introduction and elimination rules for universal and existential quantifiers; identity rules; connecting FOL proof theory to interpretations;

References

Magnus, P., Button, T., Trueman, R., & Zach, R. (2025). *Forall x: Calgary: An introduction to formal logic*. University of Calgary. https://openlogicproject.org/

Smith, P. (2020). *An introduction to formal logic* (2nd). Independently published.